

END-SEMESTER EXAMINATION, NOV-DEC, 2025

Course Code: CBCPC501

Course Title: Principle of Compiler Construction

Duration: 03 Hours

Max. Marks: 50

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	Explain the evolution of programming languages from machine language to high-level languages. How has this influenced compiler design?	5	CO1
1b	Describe the phases of a compiler with a neat block diagram. Explain the role of symbol tables and how they are used across different phases of a compiler?	5	CO1
1c	Write a short note on the following: (i) Language Processor (ii) Transition diagram (iii) Bootstrapping (iv) Just-in-time Compiler (v) Compiler construction tools	5	CO1
Q 2	Attempt any 2 parts of the following		
2a	Describe the role of a Lexical Analyser Generator (Lex/Flex). Describe the process of lexical error handling with examples.	5	CO2
2b	Given the input stream int a = b + 25; and buffer size 10 with sentinel technique: (i) Show the movement of pointers lexemeBegin and forward. (ii) Indicate when the second buffer is loaded.	5	CO2
2c	Explain token recognition and its integration with the symbol table and parser. Identify the lexemes that make up the tokens in the following program: char ch = 'A'; if (ch != 'Z') { ch = ch + 1; }	5	CO2
Q 3	Attempt any 2 parts of the following		
3a	Define context-free grammar (CFG). What is left recursion and how can it be eliminated for top-down parsing?	5	CO3
3b	Consider the following grammar: E → TE' E' → +TE' ε T → FT' T' → *FT' ε F → (E) id (i) Compute FIRST and FOLLOW sets. (ii) Construct the LL(1) predictive parsing table. (iii) Use the table to parse the input string: id + id * id	5	CO3
3c	Differentiate between shift-reduce parsing, SLR, CLR, and LALR parsers. For the grammar: S → CC C → cC d (i) Build the SLR parsing table. (ii) Show the shift-reduce parsing steps for the input: ccdd	5	CO3

Q 4	Attempt any 2 parts of the following	
4a	What is syntax-directed translation? Generate three-address code (TAC) using syntax-directed translation for the following expression: $a = b * (c + d) - e / f$	5
4b	What is backpatching? Explain the structure and purpose of an activation record along with its major components like return address, local variables, dynamic link, static link, etc.	5
4c	What is Control Flow Graph (CFG)? Explain the steps to find basic blocks. Consider the following program: 1: start 2: a = 0 3: i = 1 4: if i > N goto 11 5: a = a + i 6: i = i + 1 7: if i <= N goto 4 8: b = a * 2 9: if b > 100 goto 12 10: print b 11: end 12: print "big" 13: end2 (i) Partition into basic blocks and draw the CFG. (ii) Identify total edges and natural loops.	5
Q 5	Attempt any 2 parts of the following	
5a	Explain these optimization techniques: constant folding, dead code elimination and algebraic simplification. Given instruction sequence (RISC), each instruction has the following cycle cost: ADD=1, MUL=5, MOV=1, LOAD=2, STORE=2. Sequence: 1: LOAD R1, [A] 2: MOV R2, R1 3: ADD R2, R2, #0 4: MUL R3, R2, #4 5: STORE [B], R3 (i) Identify peephole optimizations and produce optimized sequences. (ii) Compute cycle savings.	
5b	Explain these loop optimization techniques: loop invariant code motion, loop unrolling, and loop fusion. Consider the following three-address code: 1: x = y + z 2: t = x * 2 3: u = 10 4: v = u + 5 5: return v (i) Identify which statements are dead. (ii) Provide optimized sequence.	
5c	Explain register allocation and register assignment using graph coloring or any suitable technique. Consider the following variables, show the register allocation process step by step. t1 = a + b t2 = t1 * c t3 = t1 + d t4 = t2 - t3	

Degree: B.TECH Semester: 5th
END-SEMESTER EXAMINATION, NOVEMBER 2025

Course Title: Data Mining
 Course Code: CBCPC503

Duration: 03 Hours

Max. Marks: 50

Note: - Attempt all questions in the given order only. Missing data/information (if any) may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	Explain the following terms: i. OLAP ii. Machine Learning	2.5*2=5	CO1
1b	What do you mean by data mining and also explain its goals and applications?	2+3	CO1
1c	Explain all the stages of data mining process in detail.	5	CO2
Q 2	Attempt any 2 parts of the following		
2a	What do you mean by data reduction? Explain any three methods for data reduction.	2+3	CO2
2b	What do you mean by data warehouse? Explain with its architecture?	2.5+2.5	CO2
2c	Explain the following terms: i. Data Compression ii. Data Sampling iii. Dimension Reduction	1.5+1.5+2	CO3
Q 3	Attempt any 2 parts of the following		

3a	How can you represent the input data in data mining? Explain in detail.	5	CO3
3b	What do you mean by data preprocessing and why it is so important, Explain all steps of data preprocessing	3+2	CO3
3c	Explain the techniques for effective reporting and effective data visualization.	2.5+2.5	CO4
Q 4	Attempt any 2 parts of the following		
4a	Explain the concept of Bayesian Networks with an example.	5	CO3
4b	Explain the bagging, Stacking and Boosting models.	1.5+1.5+2	CO3
4c	Define the following: i. Decision Tress ii. Covering Rules	2.5+2.5	CO4
Q 5	Attempt any 2 parts of the following		
5a	Explain the preprocessing data from medical domain with the help of an example.	5	CO5
5b	What do you mean by clustering and also explain the concept of K-means clustering?	2+3	CO5
5c	Explain in details any five data mining software	5	CO5

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